

Teacher:	Ray Joshua Balingkit	**Subject:**	Computing
Week:	5 (Term 1)	**Year:**	6
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will learn to model a robot's input, processing and output and test its response to an obstacle. They will also learn to design a useful robot with suitable sensors, actuators and a simple control rule. Through teacher demonstrations, discussion and practical activities, students will practise these skills and explain their ideas.

End-of-Week Assessment/Check: How will you know they got it?

Use teacher observation, pupil explanations and the completed practical or paper activity. Ask each pupil to explain one complete input-process-output chain and how a wrong reading could affect it. Assess whether the parts fit the task and whether pupils can explain a test result and a human safety check. Note pupils who need further guided practice.

LESSON SLOTS - WEDNESDAY AND THURSDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	23 September 2026	By the end of this lesson, students will be able to model a robot's input, processing and output and test its response to an obstacle.	**H:** Display a robot with a missing controller label. **ID:** Review the sensor, controller, actuator and feedback vocabulary. **WD:** Demonstrate a paper robot: obstacle card enters, an if-obstacle rule is applied and an arrow changes direction. **YD:** Teams run the model for clear path, obstacle and uncertain sensor readings, recording the resulting action. **C:** Ask each pupil to explain one complete input-process-output chain and how a wrong reading could affect it.	**Support:** Provide labelled robot diagrams, vocabulary prompts and control-loop cards. Accept a spoken explanation before writing; rehearse one example together. **Extension:** Ask pupils to test an unusual input, explain a possible failure and improve the design without changing its main purpose.
2	24 September 2026	By the end of this lesson, students will be able to design a useful robot with suitable sensors, actuators and a simple control rule.	**H:** Ask what small, specific classroom task a robot could help with. **ID:** Set the design brief: identify a user, task, input, action and a safe stopping condition.	**Support:** Provide labelled robot diagrams, vocabulary prompts and control-loop cards. Accept a spoken explanation before writing; rehearse one example together.



			WD: Sketch a classroom floor cleaner with an obstacle sensor, wheel motors and a stop-or-turn rule. **YD:** Pupils draw and label their design, write its control rule and test it with a partner's scenario. **C:** Assess whether the parts fit the task and whether pupils can explain a test result and a human safety check.	**Extension:** Ask pupils to test an unusual input, explain a possible failure and improve the design without changing its main purpose.
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Oxford International Primary Computing Student Book 6 printed pages 6-17
- Teacher computer and display
- Classroom computers or laptops and the class save folder
- Prepared robot pictures, labelled component diagrams and control-loop cards
- Whiteboard and markers
- Student notebooks, pencils and task sheets
- Prepared examples and the lesson success checklist



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?